

Derivation of the Adjoint Model for the Standard Lorenz-96 System

This document derives the adjoint model for the standard Lorenz-96 system. It integrates the forward model, tangent-linear model (TLM), and Jacobian entries provided in the original problem statement with the formal steps of adjoint derivation.

1 Forward Model and Tangent-Linear Model (TLM)

The standard Lorenz-96 system is defined by the following set of ordinary differential equations:

$$\frac{dx_k}{dt} = x_{k-1}(x_{k+1} - x_{k-2}) - x_k + F, \quad k = 1, \dots, N, \quad (1)$$

with cyclic indices $x_{k+N} = x_k$.

To obtain the tangent-linear model (TLM), we linearize the system about a nonlinear trajectory $x_k(t)$. Let δx_k be the perturbation. The resulting TLM is:

$$\frac{d\delta x_k}{dt} = (x_{k+1} - x_{k-2})\delta x_{k-1} + x_{k-1}\delta x_{k+1} - x_{k-1}\delta x_{k-2} - \delta x_k. \quad (2)$$

Equivalently, in vector form, the TLM is written as:

$$\frac{d\delta \mathbf{x}}{dt} = \mathbf{M}(t)\delta \mathbf{x}, \quad \delta \mathbf{x} = \begin{pmatrix} \delta x_1 \\ \vdots \\ \delta x_N \end{pmatrix}, \quad (3)$$

where the Jacobian matrix $\mathbf{M}(t)$ has the following non-zero entries (all indices are modulo N):

$$M_{k,k-1} = x_{k+1} - x_{k-2}, \quad (4)$$

$$M_{k,k+1} = x_{k-1}, \quad (5)$$

$$M_{k,k-2} = -x_{k-1}, \quad (6)$$

$$M_{k,k} = -1. \quad (7)$$

2 Adjoint Model Formulation

The adjoint model describes the evolution of the adjoint state vector $\delta \mathbf{x}^*$. In continuous time, the adjoint equation is defined using the transpose of the Jacobian matrix, integrated backward in time:

$$-\frac{d\delta \mathbf{x}^*}{dt} = \mathbf{M}^T(t)\delta \mathbf{x}^*, \quad (8)$$

or equivalently:

$$\frac{d\delta \mathbf{x}^*}{dt} = -\mathbf{M}^T(t)\delta \mathbf{x}^*. \quad (9)$$

3 Deriving the Transposed Jacobian $\mathbf{M}^T(t)$

To find the adjoint equations in component form, we need the entries of the transposed Jacobian $\mathbf{M}^T$. By definition, $(M^T)_{i,j} = M_{j,i}$. We find the non-zero entries of $\mathbf{M}^T$ for a generic row k (which corresponds to column k of $\mathbf{M}$):

$$\begin{aligned} \text{From } M_{k+1,k} = x_{k+2} - x_{k-1}, \quad & \text{we get } (M^T)_{k,k+1} = x_{k+2} - x_{k-1}, \\ \text{From } M_{k-1,k} = x_{k-2}, \quad & \text{we get } (M^T)_{k,k-1} = x_{k-2}, \\ \text{From } M_{k+2,k} = -x_{k+1}, \quad & \text{we get } (M^T)_{k,k+2} = -x_{k+1}, \\ \text{From } M_{k,k} = -1, \quad & \text{we get } (M^T)_{k,k} = -1. \end{aligned}$$

4 Component Form of the Adjoint Equations

Using the definition of the adjoint equation $-\frac{d\delta x_k^*}{dt} = \sum_j (M^T)_{k,j} \delta x_j^*$, we substitute the non-zero entries of $\mathbf{M}^T$:

$$-\frac{d\delta x_k^*}{dt} = (M^T)_{k,k+1} \delta x_{k+1}^* + (M^T)_{k,k-1} \delta x_{k-1}^* + (M^T)_{k,k+2} \delta x_{k+2}^* + (M^T)_{k,k} \delta x_k^*. \quad (10)$$

Substituting the values derived in Section 3:

$$-\frac{d\delta x_k^*}{dt} = (x_{k+2} - x_{k-1}) \delta x_{k+1}^* + x_{k-2} \delta x_{k-1}^* - x_{k+1} \delta x_{k+2}^* - \delta x_k^*. \quad (11)$$

5 Final Adjoint Model

Finally, multiply the entire equation by -1 to isolate the time derivative, and rearrange the terms to clearly show the adjoint dynamics:

$$\boxed{\frac{d\delta x_k^*}{dt} = -x_{k-2} \delta x_{k-1}^* + (x_{k-1} - x_{k+2}) \delta x_{k+1}^* + x_{k+1} \delta x_{k+2}^* + \delta x_k^*} \quad (12)$$

with cyclic indices $\delta x_{k+N}^* = \delta x_k^*$ and $x_{k+N} = x_k$.

Important Integration Note: While the tangent-linear model is integrated forward in time (from $t = 0$ to $t = T$), the adjoint model must be integrated **backward in time** (from $t = T$ to $t = 0$). The “initial” condition for this backward integration is the adjoint state at the final time T , typically derived from a cost function gradient.